HW 3 Veronica Gebbia 663838135
Ch 11A.1 ; 11B.2 ; 11B.4
Ch 12B.4

11A.1 | max temp of 3A.1 ans ≈ 217°C
$k = 4.0\times10^{-4}$ cal/s·cm·C ; $T_m = 200°C$; r = 4000 rpm
3A.1: d = 2.00 in. ; $\mu = 200$ cp
fully develop Couette flow w/ temp vary only gap y
then energy balance → $= 0 = k\left(\frac{d^2T}{dy^2}\right) + \mu\left(\frac{du}{dy}\right)^2 \rightarrow$
$k\left(\frac{d^2T}{dx^2}\right) = -\mu\left(\frac{du}{dy}\right)^2$
Couette velocity profile: $u(y) = (V/b)y \rightarrow du/dy = V/b$
$(du/dy)^2 = (V/b)^2 \rightarrow k\left(\frac{d^2T}{dy^2}\right) = -\mu\left(\frac{V}{b}\right)^2 \rightarrow$
$\frac{d^2T}{dy^2} = \frac{-\mu}{k}\left(\frac{V}{b}\right)^2$
$\frac{dT}{dy} = \frac{-\mu}{k}\left(\frac{V}{b}\right)^2 y + C_1 \rightarrow T(y) = \frac{-\mu}{2k}\left(\frac{V}{b}\right)^2 y^2 + C_1 y + C_2$
at y = 0 and y = b
$y = 0$; $T(0) = T_w \rightarrow T_w = 0 + 0 + C_2 \rightarrow T_w = C_2$
$y = b$; $T(b) = T_w \rightarrow T_w = \frac{-\mu}{2k}\left(\frac{V}{b}\right)^2 b^2 + C_1 b + T_w \rightarrow$

$C_1 = \frac{\mu V^2}{2kb} \rightarrow T(y) = T_w - \frac{\mu}{2k}\left(\frac{V}{b}\right)^2 y^2 + \frac{\mu V^2}{2kb} y \rightarrow$
$T(y) = T_w + \frac{\mu V^2}{2k}\left(\frac{y}{b} - \left(\frac{y}{b}\right)^2\right)$ $m = y/b \rightarrow$
$T(y) = T_w + \frac{\mu V^2}{2k}(m - m^2)$
max is $dT/dm = 0$ so $d/dm\,(m - m^2) = 1 - 2m = 0 \rightarrow$
$m = 1/2$ so max @ mid-gap $y = b/2$
$(1/2 - (1/2)^2) = 1/4 \rightarrow T_{max} = T_w + \mu V^2/2k\,(1/4) \rightarrow$

$T_{max} = T_w + \mu v^2/8k \rightarrow T_{max} - T_w = 1/8(\mu v^2/k)$

$v = \Omega R \rightarrow T_{max} - T_w = 1/8\left(\frac{\mu \Omega^2 R^2}{k}\right)$

$\Omega = N\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)\left(\frac{1 \text{ min}}{60 s}\right) \rightarrow$

$\Omega = 418.88 \text{ rad/s}$

$v = \Omega R \rightarrow 418.88 \text{ rad/s}(0.0254 \text{ m}) = 10.64 \text{ m/s}$

$1 \text{ cPa} = 10^{-3} \text{ Pa}\cdot\text{s} \rightarrow \mu = 200 \text{ cP} = 200 \times 10^{-3} = 0.20 \text{ Pa}\cdot\text{s}$

$k = 4.0 \times 10^{-4} \text{ cal/s}\cdot\text{cm}\cdot\text{C} \rightarrow 4.0 \times 10^{-4}\left(\frac{4.184 J}{1 \text{ cal}}\right)\left(\frac{100}{1}\right)\frac{J}{s\cdot m\cdot K}$

$k = 0.17 \text{ W/m}\cdot\text{K}$

$\Delta T_{max} = 1/8(\mu v^2/k) \rightarrow \frac{1}{8}\left(\frac{0.20 \text{ Pa}\cdot\text{s}(10.64 \text{ m/s})^2}{0.17 \text{ W/m}\cdot\text{K}}\right)$

$\Delta T_{max} \approx 16.64\,°C$

$T_{max} = \Delta T_{max} + T_w \rightarrow 200 + 16.64\,°C = 216.64\,°C$

$T_{max} \approx 216.64\,°C$

11B.2

laminar Newtonian flow in a circular tube w/ r

velocity profile: $v_z(r) = v_{z,max}(1 - (r/R)^2)$

temp dep. r & z: $T = T(r,z)$

a) $\rho C_p(v \cdot \nabla T) = k\nabla^2 T + \Phi$

for a devel. tube flow $\rightarrow v_z = v_z(r)$; $v_r = 0$; $v_\theta = 0$

$v \cdot \nabla T = v_z(\partial T/\partial z)$

$\nabla^2 T = 1/r(\partial/\partial r)(r(\partial T/\partial r))$

$\rho C_p(v_z(\partial T/\partial z)) = k(1/r(\partial/\partial r)(r(\partial T/\partial r))) \rightarrow$

$\rho C_p v_{z,max}(1 - (r/R)^2)\partial T/\partial z$

$\Phi = \mu(dv_z/dr)^2$

$\hookrightarrow d/dr(v_{z,max}(1 - r^2/R^2)) = \left(\frac{-2v_{z,max} r}{R^2}\right)^2 \rightarrow$

$= \frac{4v_{z,max}^2 r^2}{R^4} \rightarrow$

$\Phi = \frac{4\mu v_{z,max}^2 r^2}{R^4} \rightarrow$

combine →

$$\rho C_p V_{z,max}\left(1-\left(r/R\right)^2\right)\partial T/\partial z =$$

$$k\left(1/r\left(\partial/\partial r\left(r\,\partial T/\partial r\right)\right)\right) + \frac{4\mu V_{z,max}^2 r^2}{R^4}$$

restrictions: $\partial T/\partial r = 0$ at $r=0$

bound at wall $r=R \rightarrow T=T_0$; $q_r = -k\,\partial T/\partial r = 0$

b) big $z \rightarrow \partial T/\partial z = 0$ so eqn becomes →

$$0 = k\left(1/r\,\partial/\partial r\left(r\,\partial T/\partial r\right)\right) + \frac{4\mu V_{z,max}^2 r^2}{R^4} \rightarrow$$

$$1/r\,d/dr\left(r\,dT/dr\right) = \frac{-4\mu V_{z,max}^2 r^2}{kR^4} \rightarrow$$

$$\int d/dr\left(r\,dT/dr\right) = \int \frac{-4\mu V_{z,max}^2 r^3}{kR^4} \rightarrow$$

$$\int r\,dT/dr = \int \frac{-\mu V_{z,max}^2 r^3}{kR^4} + C_1/r \rightarrow$$

$$T(r) = \frac{-\mu V_{z,max}^2 r^4}{4kR^4} + C_1 \ln r + C_2$$

boundary conditions: $r=0$; $\ln r \rightarrow -\infty$ as $r \rightarrow 0$ so $C_1 = 0$

$$T(r) = \frac{-\mu V_{z,max}^2 r^4}{4kR^4} + C_2$$

wall condition $T(R) = T_0$

$$T_0 = \frac{-\mu V_{z,max}^2 R^4}{4kR^4} + C_2 \rightarrow = \frac{-\mu V_{z,max}^2}{4k} + C_2$$

$$C_2 = T_0 + \frac{\mu V_{z,max}^2}{4k}$$

$$T - T_0 = \frac{\mu V_{z,max}^2}{4k}\left(1-\left(\frac{r}{R}\right)^4\right)$$

c) adiabatic wall : $q_r(R,z)=0$; $\partial T/\partial r = 0$

$\xi = r/R \quad (0 \le \xi \le 1) \quad r = R\xi \Rightarrow dr = R\,d\xi$

$1/r \; \partial/\partial r \,(r\, \partial T/\partial r)$

$\hookrightarrow 1/R \; \partial T/\partial \xi \rightarrow r(1/R \; \partial T/\partial \xi) \rightarrow$

$(R\xi)(1/R \; \partial T/\partial \xi) = \xi \, \partial T/\partial \xi \rightarrow$

$\partial/\partial r(\xi \, \partial T/\partial \xi) = 1/R \; \partial/\partial \xi \,(\xi \, \partial T/\xi) \div r = R\xi$

$1/r \; \partial/\partial r\,(r\, \partial T/\partial r) = 1/R\xi \,(1/R \, \partial/\partial \xi\,(\xi\, \partial T/\partial \xi)) \rightarrow$

$= 1/R^2 \,(1/\xi)\, \partial/\partial \xi\,(\xi \, \partial T/\partial \xi)$

combine $\rightarrow$

$$\rho \hat{C}_p v_{z,max}(1-\xi)^2 \, \partial T/\partial z = k/R^2 \; 1/\xi \; \partial/\partial \xi\,(\xi\, \partial T/\partial \xi) + \frac{4\mu v_{z,max}^2}{R^2}\xi^2$$

11B.4

N.B : $\theta = \theta_1$ @ $T = T_1$

S.B : $\theta = \pi - \theta_1$ @ $T = T_2$

$\nabla^2 T = 0$

$\hookrightarrow = 1/r^2 \; \partial/\partial r\,(r^2\, \partial T/\partial r) + 1/r^2 \sin\theta \; \partial/\partial\theta\,(\sin\theta \, \partial T/\partial \theta) + \ldots$

$1/r^2 \sin^2\theta \,(\partial^2 T/\partial \phi^2)$

$\partial T/\partial \phi = 0 \rightarrow \partial^2 T/\partial \phi^2 = 0$

$q_r = -k\, \partial T/\partial r = 0$ at $r = R_1$ and $r = R_2$

$\partial T/\partial r = 0$ T don't depend on r $\Rightarrow \partial T/\partial r = 0$

$1/\sin\theta \; d/d\theta\,(\sin\theta \, dT/d\theta) = 0$

$d/d\theta\,(\sin\theta\, dT/d\theta) = 0$

$\hookrightarrow = C_1$

$dT/d\theta = C_1/\sin\theta$

$T(\theta) = C_1 \int \csc\theta \, d\theta + C_2$

$\hookrightarrow \ln|\tan \theta/2| + \text{const.}$

$T(\theta) = C_1 \ln(\tan\theta/2) + C_2$

$T(\theta_1) = T_1 \qquad T(\pi - \theta_1) = T_2$

$T_1 = C_1 \ln(\tan \theta_1/2) + C_2$

12B.4

a) z-mom: $\rho\left(\frac{\partial v_z}{\partial t} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right) + \rho g$

stead → $\partial/\partial t = 0$; full devel → $\partial v_z/\partial z = 0$; no direct → $v_y \approx 0, v_z(y)$

press. ∇ in z → $\partial p/\partial z \approx 0$; no vary in z → $\partial^2 v_z/\partial z^2 = 0$

reduced to → $0 = \mu\frac{d^2 v_z}{dy^2} + \rho g$

bounds: no slip → $v_z(0) = 0$; 0 shear @ free surf → $dv_z/dy = 0$

$\int d^2v_z/dy^2 = \int -\rho g/\mu \rightarrow dv_z/dy = (-\rho g/\mu)y + C_1$

- @ $y = \delta \rightarrow 0 = (-\rho g/\mu)\delta + C_1 \rightarrow C_1 = (\rho g/\mu)\delta$

$\int dv_z/dy = \int(\rho g/\mu)(\delta - y) \rightarrow v_z = (\rho g/\mu)(\delta y - y^2/2) + C_2 \rightarrow$

- @ $v_z(0) = 0 \rightarrow C_2 = 0$

$v_z(y) = (\rho g/\mu)(\delta y - y^2/2)$

- N.D:

$v_z(y) = \rho g/\mu\,(\delta^2/2)(2y/\delta - (y/\delta)^2) \rightarrow$

$v_{z,max} = \frac{\rho g\delta^2}{2\mu}$

$v_z = v_{z,max}(2(y/\delta) - (y/\delta)^2)$

- near wall: $y < \delta \rightarrow (y/\delta)^2 < y/\delta$

$v_z = v_{z,max}(2(y/\delta) - (y/\delta)^2) \rightarrow \approx v_{z,max}(2y/\delta) \rightarrow$

$v_{z,max} = \frac{\rho g\delta^2}{2\mu} \rightarrow v_z = 2\left(\frac{\rho g\delta^2}{2\mu}\right)\frac{y}{\delta} \rightarrow v_z = \frac{\rho g\delta}{\mu}y$

b) stead E eqn: $\partial T/\partial t = 0$; k, ρ, C_p = const. ; $q_m = 0$; $\Phi \approx 0$

; $-\beta_T T\, Dp/Dt \approx 0 \rightarrow \rho\hat{C}_p(v\nabla T) = k\nabla^2 T$

↳ $\underbrace{v_y\,\partial T/\partial y}_{=0} + v_z(\partial T/\partial z) \rightarrow v_z(\partial T/\partial z)$

$\nabla^2 T = \partial^2 T/\partial y^2 + \partial^2 T/\partial z^2$ $\quad l_T < L_z$ so $\partial^2 T/\partial z^2 < \partial^2 T/\partial y^2$

↳ $= \partial^2 T/y^2$ $\quad$ ↳ temp Δ near wall

$\rho\hat{C}_p v_z\,\partial T/\partial z = k\,\partial^2 T/\partial y^2$ $\quad$ w/ $v_z = (\rho g\delta/\mu)y \rightarrow$

$\rho \hat{C}_p (\rho g \delta/\mu) y \cdot \partial T/\partial z = k\, \partial^2 T/\partial y^2$

$y\, \partial T/\partial z = \dfrac{\mu k}{\rho^2 \hat{C}_p g \delta} \dfrac{\partial^2 T}{\partial y^2}$; $\beta = \dfrac{\mu k}{\rho^2 \hat{C}_p g \delta}$

so $y \dfrac{\partial T}{\partial z} = \beta \dfrac{\partial^2 T}{\partial y^2}$

c) B.C 1: $T = T_0$ for $z = 0$; $y > 0$

$T(y, z=0) = T_0$ for $0 < y \leq \delta$

B.C 2: $T = T_0$ for $y \to \infty$, z finite

$y\, \partial T/\partial z = \beta\, \partial^2 T/\partial y^2 \quad \beta = \mu k/\rho^2 \hat{C}_p g \delta$

T changes only for $0 \leq y \leq l(z) \to$ heat part

$y > l(z)$, $T \approx T_0$

$\partial T/\partial z \sim \Delta T/z$; $\partial^2 T/\partial y^2 \sim \Delta T/l^2$; $y \sim l$

$l(\Delta T/z) \sim \beta(\Delta T/l^2) \to l^3 \sim \beta z \to l_z \sim (\beta z)^{1/3}$

short contact time : $l(z) < \delta \to (\beta z)^{1/3} < \delta$

$y = \delta$ so $T(\delta, z) \approx T_0$

since fluid near wall $(y < l)$ is unaffected by anything far $(y \sim \delta)$ then $y = \delta$ to $\infty = y \to$

$\lim_{y \to \infty} T(y, z) = T_0$ for finite z

B.C 3: $T = T_1$ for $y = 0$; $z > 0$

$T(0, z) = T_1$ $(z > 0)$ since the wall is at a const. temp (T_1) and the fluid temp at the wall = wall temp.

d) $\Theta(\eta) = T - T_0/T_1 - T_0$ bound vals to 0/1 so

$T = T_0 + (T_1 - T_0)\Theta$

$\eta = \dfrac{y}{(9\beta z)^{1/3}}$; $\Theta = \Theta(\eta)$ so $\partial\Theta/\partial z = d\Theta/d\eta\, \partial\eta/dz$;

$\partial\Theta/\partial y = d\Theta/d\eta\, \partial\eta/dy$

$\to (9\beta z)^{-1/3}$

$\eta = y(9\beta)^{-1/3} z^{-1/3} \to$

$\partial\eta/\partial z = y(9\beta)^{-1/3}(-1/3) z^{-4/3}$; $y(9\beta)^{-1/3} z^{-1/3} = \eta \to$

$\partial\eta/\partial z = -\eta/3z$

$T = T_0 + (T_1 - T_0)\theta$

$\partial T/\partial z = (T_1 - T_0)\,\partial\theta/\partial z \rightarrow (T_1 - T_0)\, d\theta/d\eta\,(-\eta/3z)$

$\partial T/\partial y = (T_1 - T_0)\,\partial\theta/\partial y \rightarrow (T_1 - T_0)\, d\theta/d\eta\,(9\beta z)^{-1/3}$

$\partial^2 T/\partial y^2 = (T_1 - T_0)\,\partial/\partial y\,(d\theta/d\eta\,(9\beta z)^{-1/3}$

$\hookrightarrow = (T_1 - T_0)(9\beta z)^{-1/3}\,\partial/\partial y\,(d\theta/d\eta)$

$\hookrightarrow = d^2\theta/d\eta^2\,\partial\eta/\partial y = d^2\theta/d\eta^2\,(9\beta z)^{-1/3}$

$\partial^2 T/\partial y^2 = (T_1 - T_0)(9\beta z)^{-2/3}\, d^2\theta/d\eta^2$

left $y\,\partial T/\partial z = y\,(T_1 - T_0)\, d\theta/d\eta\,(-\eta/3z)$; $y = \eta\,(9\beta z)^{1/3} \rightarrow$

$y\,\partial T/\partial z = (T_1 - T_0)(\eta\,(9\beta z)^{1/3})(-\eta/3z)\, d\theta/d\eta \rightarrow$

$= (T_1 - T_0)(-\eta^2/3\,(9\beta)^{1/3} z^{-2/3})\, d\theta/d\eta$

right $\beta\,\partial^2 T/\partial y^2 \rightarrow \beta\,(T_1 - T_0)(9\beta z)^{-2/3}\, d^2\theta/d\eta^2 \rightarrow$

$(T_1 - T_0)(\beta(9\beta)^{-2/3} z^{-2/3})\, d^2\theta/d\eta^2$ w/ remain const. $\rightarrow$

$d^2\theta/d\eta^2 + 3\eta^2\, d\theta/d\eta = 0$

B.C : at the wall ($y=0$): $\eta = 0$; $T = T_1 \rightarrow \theta = 1$ so $\theta(0) = 1$

far from wall ($y = \infty$): $\eta \rightarrow \infty$; $T \rightarrow T_0 \rightarrow \theta \rightarrow 0$ so $\theta(\infty) = 0$

e) $d^2\theta/d\eta^2 + 3\eta^2\, d\theta/d\eta = 0$

; $p = d\theta/d\eta \rightarrow d^2\theta/d\eta^2 = dp/d\eta$

$dp/d\eta + 3\eta^2 p = 0 \rightarrow \int 1/p\, dp = \int -3\eta^2\, d\eta \rightarrow$

$\ln p = -\eta^3 + \ln C_1 \rightarrow p = C_1 e^{-\eta^3}$ so $\int d\theta/d\eta = \int C_1 e^{-\eta^3}$

$\theta(\eta) - \theta(0) = C_1 \int_0^{\eta} e^{-m^3}\, dm$; $\theta(0) = 1 \rightarrow$

$\theta(\eta) = 1 + C_1 \int_0^{\eta} e^{-m^3}\, dm$; $\theta(\infty) = 0 \rightarrow$

$0 = 1 + C_1 \int_0^{\infty} e^{-m^3}\, dm \rightarrow C_1 = -\int_0^{\infty} e^{-m^3}\, dm \rightarrow$

$$\theta(\eta) = 1 - \frac{\int_0^{\eta} e^{-m^3}\, dm}{\int_0^{\infty} e^{-m^3}\, dm} \rightarrow = \frac{\int_{\eta}^{\infty} e^{-m^3}\, dm}{\underbrace{\int_0^{\infty} e^{-m^3}\, dm}_{\Gamma(4/3)}} \rightarrow$$

$$\theta(\eta) = \frac{1}{\Gamma(4/3)} \int_{\eta}^{\infty} e^{-\eta^3}\, d\eta$$

f) heat flux (+) : $q(z) = -k\,\partial T/\partial y$

$T = T_0 + (T_1 - T_0)\theta \rightarrow \partial T/\partial y = (T_1 - T_0)\,\partial\theta/\partial y$

$\hookrightarrow d\theta/d\eta\;\partial\eta/\partial y$

$\hookrightarrow (9\beta z)^{-1/3}$

$\partial T/\partial y|_{y=0} \rightarrow = (T_1 - T_0)\,d\theta/d\eta)|_{\eta=0}\,(9\beta z)^{-1/3}$

$\theta(\eta) = 1/\Gamma(4/3)\int_\eta^\infty e^{-m^3}dm \rightarrow d\theta/d\eta = -1/\Gamma(4/3)\,e^{-\eta^3}$

at $\eta = 0 \rightarrow d\theta/d\eta|_0 = -1/\Gamma(4/3)$

$q(z)|_{y=0} = -k(T_1 - T_0)(-1/\Gamma(4/3))(9\beta z^{-1/3})$

$0 < z < L$

$q_{avg}|_{y=0} = 1/L\int_0^L q(z)\,dz \rightarrow$

$\frac{k(T_1 - T_0)}{\Gamma(4/3)}(9\beta)^{-1/3}\;1/L\int_0^L z^{-1/3}dz \rightarrow 3/2\,z^{2/3}]_0^L \rightarrow 3/2\,z^{2/3} \rightarrow$

$q_{avg}|_{y=0} = \frac{k(T_1 - T_0)}{\Gamma(4/3)}(9\beta)^{-1/3}\cdot 1/L\,(3/2\,L^{2/3}) \rightarrow$

$= 3/2\,\frac{k(T_1 - T_0)}{\Gamma(4/3)}(9\beta L)^{-1/3}$ so

$q_{avg}|_{y=0} = 3/2\,\frac{(9\beta L)^{-1/3}}{\Gamma(4/3)}\,k(T_1 - T_0)$

#5) $\rho C_p\,\partial T/\partial t = k\nabla^2 T \quad \alpha = k/\rho C_p$ so

$\hookrightarrow \alpha\nabla^2 T$

(r, θ, ϕ) if $T = T(r,t)$ so

$\nabla^2 T = 1/r^2\;\partial/\partial r\,(r^2\,\partial T/\partial r)$

1- $\partial T/\partial t = \alpha\,(1/r^2\;\partial/\partial r\,(r^2\,\partial T/\partial r)) \quad 0 < r < R \quad t > 0$

I.C: $T(r, 0) = T_0 \quad 0 \le r \le R$

S.B: surface set to T_1 for $t > 0 \rightarrow T(R, t) = T_1$

$t > 0$

center symm: $\partial T/\partial r = 0$

2- $\theta = T_1 - T / T_1 - T_0$; $\eta = r/R$; $\tau = \alpha t / R^2$

$\theta = T_1 - T / T_1 - T_0 \rightarrow T = T_1 - (T_1 - T_0)\theta$ diff w/ rspt to $t \rightarrow$

$\partial T/\partial t = -(T_1 - T_0)\partial\theta/\partial t$; $\tau \rightarrow t = \tau R^2/\alpha \rightarrow$

$\partial\theta/\partial t = \partial\theta/\partial\tau \, \partial\tau/\partial t \rightarrow = \partial\theta/\partial\tau \, \alpha/R^2 \rightarrow$

$\partial T/\partial t = -(T_1 - T_0)\alpha/R^2 \, \partial\theta/\partial\tau$ w/ $r = R\eta \rightarrow \partial/\partial r = 1/R \, \partial/\partial\eta$

$\partial T/\partial r = -(T_1 - T_0) 1/R \, \partial\theta/\partial\eta$

w/ $\partial T/\partial t = \alpha(1/r^2)\partial/\partial r(r^2 \partial T/\partial r)$; $r = R\eta$;

$\partial T/\partial t = -(T_1 - T_0)\alpha/R^2 \, \partial\theta/\partial\tau$; $\partial T/\partial r = -(T_1 - T_0) 1/R \, \partial\theta/\partial\eta$

$= -(T_1 - T_0)\alpha/R^2 \, \partial\theta/\partial\tau$

$$\alpha\left(\frac{1}{(R\eta)^2}\right)\frac{\partial}{\partial r}\left((R\eta)^2\cdot(-(T_1-T_0)1/R\,\partial\theta/\partial\eta))\right) \rightarrow$$

$$= \alpha \, \frac{1}{R^2\eta^2} \, \frac{\partial}{\partial r}\left(-(T_1 - T_0)R\eta^2\,\partial\theta/\partial\eta)\right) \rightarrow$$

$= -(T_1 - T_0)\alpha/R^2 (1/\eta^2)\partial/\partial\eta(\eta^2 \partial\theta/\partial\eta) \rightarrow$

$\partial\theta/\partial\tau = (1/\eta^2)\partial/\partial\eta(\eta^2\partial\theta/\partial\eta)$ $0 < \eta < 1$, $\tau > 0$

I.C : $T = T_0 \rightarrow \theta = 1 \rightarrow \theta(\eta, 0) = 1$

surf. : $T(R, t) = T_1 \rightarrow \theta(1, \tau) = 0 \rightarrow \theta(1, \tau) = 0$

center symm: $\partial T/\partial r = 0 \rightarrow \partial\theta/\partial\eta = 0$ at $\eta = 0 \rightarrow$

$\partial\theta/\partial\eta|_{\eta=0} = 0$

3- $\partial\theta/\partial\tau = (1/\eta^2)\partial/\partial\eta(\eta^2\partial\theta/\partial\eta)$

$\theta(\eta, \tau) = F(\eta)G(\tau) \rightarrow F(\eta)G'(\tau) = (1/\eta^2) d/d\eta(\eta^2 F'(\eta))G(\tau)$

$G'/G = 1/F (1/\eta^2) d/d\eta(\eta^2 F') = -\lambda^2$

t ODE - $G' + \lambda^2 G = 0 \rightarrow G(\tau) = e^{-\lambda^2\tau}$

r ODE - $(1/\eta^2) d/d\eta(\eta^2 F') + \lambda^2 F = 0$

let $u(\eta) = \eta F(\eta) \rightarrow u'' + \lambda^2 u = 0$; $u = A\sin(\lambda\eta) + \ldots$

$$B\cos(\lambda\eta) \rightarrow F(\eta) = u/\eta \rightarrow \frac{A\sin(\lambda\eta) + B\cos(\lambda\eta)}{\eta}$$

at $\eta = 0 \rightarrow \eta \rightarrow 0$ so $\cos(\lambda\eta) \rightarrow 1$ so $B\cos(\lambda\eta)/\eta$

$$\text{since } B = 0 \text{ X so } F(\eta) = \frac{A\sin(\lambda\eta)}{\eta}$$

$B.C: \theta(1,\tau)=0 \rightarrow F(1)=0 \rightarrow F(1)=A\sin(\lambda)=0 \rightarrow$
$\sin(\lambda)=0 \rightarrow \lambda=n\pi$ functi: $F_n(\eta)=\dfrac{\sin(n\pi\eta)}{\eta}$
$\because G_n(\tau)=e^{-(n\pi)^2\tau}$

gen soln: $\theta(\eta,\tau)=\sum_{n=1}^{\infty} A_n\left(\dfrac{\sin n\pi\eta}{\eta}\right)e^{-(n\pi)^2\tau}$

at $\tau=0: 1=\sum_{n=1}^{\infty} A_n\left(\dfrac{\sin n\pi\eta}{\eta}\right) \rightarrow$

$\eta=\sum_{n=1}^{\infty} A_n\sin(n\pi\eta)$ sine series $\rightarrow$

$A_n=2\int_0^1 \eta\sin(n\pi\eta)\,d\eta \rightarrow$

$-\eta\left(\dfrac{\cos(n\pi\eta)}{n\pi}\right)\Big|_0^1+\int_0^1 \dfrac{\cos(n\pi\eta)}{n\pi}\,d\eta \rightarrow$

$\dfrac{-\cos(n\pi)}{n\pi}+\dfrac{1}{n\pi}\left(\dfrac{\sin(n\pi\eta)}{n\pi}\right)\Big|_0^1$

$\int_0^1 \eta\sin(n\pi\eta)\,d\eta=\dfrac{-\cos(n\pi)}{n\pi}=\dfrac{(-1)^{n+1}}{n\pi}$

so $A_n=2\left(\dfrac{-1^{n+1}}{n\pi}\right)$

$\theta(\eta,\tau)=\dfrac{2}{\eta}\sum_{n=1}^{\infty}\left(\dfrac{-1^{n+1}}{n\pi}\right)\sin(n\pi\eta)e^{-(n\pi)^2\tau}$

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